VO2 Master Analyzer Bluetooth Communication

V4.3.0

September 13, 2022

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Summary

This document describes how to communicate with the VO2 Master Analyzer directly over Bluetooth. Attached to this summary is code for using the VO2 Master Analyzer in your third-party application. Please contact stephen.itodyck@vo2master.com should you have any questions, comments, or concerns.

List of VO2 Master Unit Device Model Numbers

Device Model Number	Release Date
1.3.0	December 18, 2019
1.4.0	October 18, 2020
1.4.4	May 5, 2021
1.5.0	October 7, 2021
1.5.3	February 3, 2022
1.5.6	May 25, 2022
1.6.0	August 25, 2022

Device Operation

The following section briefly describes the operating procedure of the VO2 Master Analyzer upon startup.

Upon powering on the device, it will begin calibrating its sensors. The calibration of these sensor is mandatory, and this calibration information does not persist across power cycles. The sensor calibration procedure is as follows:

1. Determine the zero-point of the flow sensor.
    a. This requires that the device is still for 10 seconds

        i. An error code is generated if there is too much noise detected i.e., external fan, very loud room
2. (Version 1.5 Only) Syringe calibration of the flow sensor
3. Oxygen sensor is calibrated to ambient air.
    a. Occurs during breathing above the minimum ventilation for a given user piece
        i. Medium has a minimum ventilation of 30L/min
    b. Using the vacuum created during inhalation the device will passively sample the ambient and use this reading as a baseline
        i. Updates are sent about the progress of the "flow delay percent" ranging from 0-100%
4. Device switches into metric recording mode and measures breath metrics

If the device is left idle for a period of 15 minutes it will shutoff and send a command over the VO2 Master Custom Service, for more information on this see the below section on the custom service.

VO2 Master Analyzer Services:

- Device Information Service (Bluetooth Standard)

  https://www.bluetooth.com/specifications/specs/device-information-service-1-1/\

- Battery Service (Bluetooth Standard)

  https://www.bluetooth.com/specifications/specs/?status=active&show_latest_version=0&show_latest_version=1&keyword=battery&filter=

- VO2 Master Service (Custom service written by VO2 Master)

More information on Bluetooth Standard services can be found here:

https://www.bluetooth.com/

The VO2 Master Manager app may be used as a template for how these communications may be implemented. Included with this document are high-resolution product images, app screenshots, app resources, Bluetooth parsing and data structure code files.

Device Information Service

UUID: 180A

Model number is used for version-specific parsing of data. It is imperative that this information is polled prior to parsing any data.

VO2MasterHandler.cs: ensure that the VO2 Master Analyzer's Device Information service has been polled prior to receiving any data. This class relies on the Model Number string being populated in order to *GetDeviceVersion*(). *GetDeviceVersion*() only parses the model number string once upon its first access. Model number are in the format "Major.Minor.Revision", I.E., "1.3.0".

Battery Service

UUID: 180F

The battery service is used to send low battery warnings. The battery level is initially set to 100. Once the battery has met the voltage threshold that is low, the value is set to 20. Once the VO2 Master Analyzer is out of battery, it transmits a diagnostic code through the custom service detailed below prior to shutting off.

VO2 Master Custom Service

UUID: 00001523-1212-EFDE-1523-785FEABCD123

The VO2 Master Analyzer uses a custom Bluetooth Service to transmit ventilatory, gas exchange, environmental, and syringe calibration data. The data transmitted in these communications are integers multiplied by a scaling factor to preserve some floating-point decimal places. If a data type has a multiplier of 100 it means that the received data needs to be divided by 100 to get the actual value of the data type.

Furthermore, this service includes a simple bi-directional communication for the application of **non-persistent** settings to the VO2 Master Analyzer, as well as the transmission diagnostic information.

If the value of FeO_2 and VO_2 for a given row are both exactly 22.0, said row is a "Ventilation-only row". Such a row contains all the data except the gas exchange parameters, which are all set to 22. See "VO2MasterMeasurementGasExchange. IsDataWithinRange()".

Version 1.3.X Read-Only Characteristics

For devices that are on any variation of the V1.3 firmware there are three characteristics the device will send data over the ventilatory, gas exchange and environment characteristics.

Ventilatory Characteristic

UUID: 00001527-1212-EFDE-1523-785FEABCD123

Here is a description of each measured trend and its byte location within the packet.

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Respiratory Frequency	100	Breaths per minute [bpm].
3, 4	Tidal Volume	100	Volume of a given breath [L]. Reported in BTPS.
5, 6	Ventilation	100	Breathing rate per minute [L/min]. Reported in BTPS.

Each pair of bytes represents an unsigned 16-bit integer.

Gas Exchange Characteristic

UUID: 00001528-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	FeO_2	100	Fraction of Expired Oxygen [%]
3, 4	Unused Data	N/A	Unused Bytes
5, 6	VO_2	1	Volume of Oxygen Consumed [mL/min]
7, 8	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Environment Characteristic

UUID: 00001529-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Ambient Pressure	10	Environmental pressure [hPa]
3, 4	Device Temperature	100	Device internal temperature [C]
5, 6	Device Humidity	100	Humidity as measured at the gas sensors [%RH]
7, 8	Ambient O_2	100	Ambient oxygen concentration [%]
9, 10	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Version 1.4.X Read-Only Characteristics

For devices that are on any variation of the V1.4 firmware there are three characteristics the device will send data over the ventilatory, gas exchange, environment, and gas calibration characteristics.

Ventilatory Characteristic

UUID: 00001527-1212-EFDE-1523-785FEABCD123

Here is a description of each measured trend and its byte location within the packet.

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Respiratory Frequency	100	Breaths per minute [bpm].
3, 4	Tidal Volume	100	Volume of a given breath [L]. Reported in BTPS.
5, 6	Ventilation	100	Breathing rate per minute [L/min]. Reported in BTPS.

Each pair of bytes represents an unsigned 16-bit integer.

Gas Exchange Characteristic

UUID: 00001528-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	FeO_2	100	Fraction of Expired Oxygen [%]
3, 4	Unused Data	N/A	Unused Bytes
5, 6	VO_2	1	Volume of Oxygen Consumed [mL/min]
7, 8	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Environment Characteristic

UUID: 00001529-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Ambient Pressure	10	Environmental pressure [hPa]
3, 4	Device Temperature	100	Device internal temperature [C]
5, 6	Device Humidity	100	Humidity as measured at the gas sensors [%RH]
7, 8	Ambient O_2	100	Ambient oxygen concentration [%]
9, 10	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Ambient Gas Calibration Characteristic

UUID: 00001531-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2, 3, 4	Calibration Time	1	Time since Unix epoch [s]
5, 6	Calibration CO_2	100	Percentage of CO_2 used for calibration [%]
7, 8	Calibration O_2	100	Percentage of O_2 used for calibration [%]
9, 10	Calibration O_2	100	Percentage of O_2 used for calibration [%]
11, 12	O_2 Coefficient	5000000	Conversion factor from raw O_2 to percent
13, 14, 15, 16 [1]	Calibration Pressure	100	Environmental pressure during calibration [hPa]
17, 18	Calibration Temperature	100	Environmental temperature during calibration [°C]
19, 20	Calibration Humidity	100	Environmental humidity during calibration [%RH]

Each quartet of bytes represents an unsigned 32-bit integer.
[1] The first two bits of calibration pressure should be discarded

Version 1.5.X Read-Only Characteristics

For devices that are on any variation of the V1.5 firmware there are three characteristics the device will send data over the ventilatory, gas exchange, environment, and gas calibration characteristics.

Ventilatory Characteristic

UUID: 00001527-1212-EFDE-1523-785FEABCD123

Here is a description of each measured trend and its byte location within the packet.

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Respiratory Frequency	100	Breaths per minute [bpm].
3, 4	Tidal Volume	100	Volume of a given breath [L]. Reported in BTPS.
5, 6	Ventilation	100	Breathing rate per minute [L/min]. Reported in BTPS.

Each pair of bytes represents an unsigned 16-bit integer.

Gas Exchange Characteristic

UUID: 00001528-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	FeO_2	100	Fraction of Expired Oxygen [%]
3, 4	Unused Data	N/A	Unused Bytes
5, 6	VO_2	6	Volume of Oxygen Consumed [mL/min]
7, 8	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Environment Characteristic

UUID: 00001529-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Ambient Pressure	10	Environmental pressure [hPa]
3, 4	Device Temperature	100	Device internal temperature [C]
5, 6	Device Humidity	100	Humidity as measured at the gas sensors [%RH]
7, 8	Ambient O_2	100	Ambient oxygen concentration [%]
9, 10	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Ambient Gas Calibration Characteristic

UUID: 00001531-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Raw O_2 Reading	1	Raw O_2 reading from sensor
3, 4	O_2 Coefficient	5000000	Conversion factor from raw O_2 to percent
5, 6	Calibration Pressure	10	Environmental pressure during calibration [hPa]
7, 8	Calibration Temperature	100	Environmental temperature during calibration [°C]
9, 10	Sensor Calibration Temperature	100	O_2 sensor temperature during calibration [°C]
11, 12	Calibration Humidity	100	Environmental humidity during calibration [%RH]

Each pair of bytes represents an unsigned 16-bit integer.

Syringe Flow Calibration Characteristic

UUID: 00001532-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Goal Volume	1000	Goal tidal volume for syringe calibration [L]
3, 4	Mean Volume	1000	Mean tidal volume from syringe calibration [L]
5, 6	Standard Deviation	1000	Standard deviation of syringe calibration strokes [L]
7, 8	Max Volume	1000	Maximum recorded tidal volume from syringe calibration [L]
9, 10	Minimum Volume	1000	Minimum recorded tidal volume from syringe calibration [L]
11, 12	Number of breaths	1	Number of valid breaths used in syringe calibration [# of breaths]
13, 14	Venturi Size	1	Venturi size that syringe calibration was performed on [*User Piece Size]

Each pair of bytes represents an unsigned 16-bit integer.

*User Piece Size is an enumeration with the user piece. See the In-depth Explanation of Important Commands subsection below for more details.

Version 1.6.X Read-Only Characteristics

For devices that are on any variation of the V1.5 firmware there are three characteristics the device will send data over the ventilatory, gas exchange, environment, and gas calibration characteristics.

Ventilatory Characteristic

UUID: 00001527-1212-EFDE-1523-785FEABCD123

Here is a description of each measured trend and its byte location within the packet.

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Respiratory Frequency	100	Breaths per minute [bpm].
3, 4	Tidal Volume	100	Volume of a given breath [L]. Reported in BTPS.
5, 6	Ventilation	100	Breathing rate per minute [L/min]. Reported in BTPS.

Each pair of bytes represents an unsigned 16-bit integer.

Gas Exchange Characteristic

UUID: 00001528-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	FeO_2	100	Fraction of Expired Oxygen [%]
3, 4	Unused Data	N/A	Unused Bytes
5, 6	VO_2	6	Volume of Oxygen Consumed [mL/min]
7, 8	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Environment Characteristic

UUID: 00001529-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Ambient Pressure	10	Environmental pressure [hPa]
3, 4	Device Temperature	100	Device internal temperature [C]
5, 6	Device Humidity	100	Humidity as measured at the gas sensors [%RH]
7, 8	Ambient O_2	100	Ambient oxygen concentration [%]
9, 10	Unused Data	N/A	Unused Bytes

Each pair of bytes represents an unsigned 16-bit integer.

Ambient Gas Calibration Characteristic

UUID: 00001531-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Raw O_2 Reading	1	Raw O_2 reading from sensor
3, 4	O_2 Coefficient	5000000	Conversion factor from raw O_2 to percent
5, 6	Calibration Pressure	10	Environmental pressure during calibration [hPa]
7, 8	Calibration Temperature	100	Environmental temperature during calibration [°C]
9, 10	Sensor Calibration Temperature	100	O_2 sensor temperature during calibration [°C]
11, 12	Calibration Humidity	100	Environmental humidity during calibration [%RH]

Each pair of bytes represents an unsigned 16-bit integer.

Syringe Flow Calibration Characteristic

UUID: 00001532-1212-EFDE-1523-785FEABCD123

Bytes	*Trend Name*	*Multiplier*	*Description*
1, 2	Goal Volume	1000	Goal tidal volume for syringe calibration [L]
3, 4	Mean Volume	1000	Mean tidal volume from syringe calibration [L]
5, 6	Standard Deviation	1000	Standard deviation of syringe calibration strokes [L]
7, 8	Max Volume	1000	Maximum recorded tidal volume from syringe calibration [L]
9, 10	Minimum Volume	1000	Minimum recorded tidal volume from syringe calibration [L]
11, 12	Number of breaths	1	Number of valid breaths used in syringe calibration [# of breaths]
13, 14	Venturi Size	1	Venturi size that syringe calibration was performed on [*User Piece Size]

Each pair of bytes represents an unsigned 16-bit integer.

*User Piece Size is an enumeration with the user piece. See the In-depth Explanation of Important Commands subsection below for more details.

COM IN / COM OUT Characteristics

COM IN UUID: 00001525-1212-EFDE-1523-785FEABCD123
COM OUT UUID: 00001526-1212-EFDE-1523-785FEABCD123

Both COM IN and COM OUT characteristics use a command/value structure for bi-directional communication. The first byte is the Command. The length of the value depends on the VO2 Master Analyzer version. The COM IN characteristic is meant for sending commands from the app to a VO2 Master Analyzer.

Every time the app connects to the VO2 Master Analyzer, the app must apply all settings. **The VO2 Master Analyzer settings <u>do not</u> persist** between power cycles. These commands are described more in-depth in the sections below.

For VO2 Master Analyzer V1.3 (Device Information Service Model Number 1.3.X):

Byte	*Description*
1	COMMAND
2	VALUE

For VO2 Master Analyzer V1.4 / V1.5 / 1.6 (Device Information Service Model Number 1.4.X / 1.5.X / 1.6.X)

Byte	*Description*
1	COMMAND
2	VALUE1
3	VALUE2

```
var value = (ushort)data[1];                    //Convert VALUE1 to byte.
value |= (ushort)(((ushort)data[2]) << 8);      //Shift VALUE2 and bit bash into VALUE1.
```

To see the possible commands, see the *VO2MasterDeviceCommandType.cs* file. Some commands only function on certain versions of the VO2 Master Analyzer. These version specific commands are described within the *VO2MasterDeviceCommandType.cs* file. The range of compatible command enumerations for each version of the VO2 Master Analyzer are listed in the below table. To maintain version compatibility, deprecated commands remain in the enumeration list but will not function on incompatible devices.

VO2 Master Version	Bluetooth Command Range
1.3	0-17
1.4	0-21
1.5	0-28
1.6	0-28

The app uses the GET commands on the COM IN characteristic to request information from the VO2 Master Analyzer. The VO2 Master Analyzer responds with the same GET command on the COM OUT characteristic accompanied by the requested value if appropriate.

If the application uses a “SET” command on the COM IN characteristic, it will modify a setting on the VO2 Master Analyzer. Subsequently, the analyzer will respond on the COM OUT characteristic with the corresponding “GET” command. For example, if the application were to send the command VM_BLE_SET_VENTURI_SIZE (command 3) on COM IN along with the value of 2, the device will respond on COM OUT with a value of 2.

In-depth Explanation of Important Commands

There are some commands that can be communicated with the VO2 Master device to modify the function of the device. If they are not communicated to the device, it will use a default value and may cause errors in the reported data. Some of these device commands are not used or are exclusive to certain versions of the VO2 Master Analyzer, these versions differences are listed in the associated command description.

Set Device State (Command #1)

The device state determines what data is being sent over the read-only characteristics and the current functionality of the device.

Enum Value	State	Ver.	State Description
0	Unused	N/A	Unused state.
1	Calibrate to Ambient Air (default)	1.5/ 1.4/ 1.3	Device is calibrating sensors to ambient air. After calibration criteria is met the device will exit this state and enter the record state. Can be re-entered to recalibrate the device. No data is sent over the read-only characteristics, but progress is communicated via the COM OUT characteristic command #5. Upon completion of the calibration, information about the calibration is sent over the Ambient Gas Calibration characteristic.
2	Record VO2 Metrics	1.5/ 1.4/ 1.3	Polls sensors to produce VO2 metrics. The device will enter this state on its own upon completion of the calibration to ambient air. Data is sent over the Ventilatory, Gas Exchange and Environmental characteristics between breaths.
3	Unused	N/A	Unused state
4	Calibrate Flow Sensor	1.5	Device is attempting to determine flow deviation from expected results. This is done per user piece and is **not** retained between power cycles. When entering this state, the device will default to calibrating with a 3L tidal volume and 10 breaths to complete calibration, which can be changed as described by command 22/24 below. **In versions 1.6.0 and greater the criteria for calibration were made more stringent and now has the additional following Rf criteria: Resting 12.5-17.5, Medium 27.5-32.5, Large 37.5-42.5.** When the calibration criteria are met data about the flow calibration will be sent over the Syringe Flow Calibration characteristic.

Set User Piece Size (Command 3)

Sets which user piece is used for the calculation of ventilation metrics. If performing a flow sensor calibration, this must be set before changing the device's state.

Enum Value	User Piece Size	Ver.	User Piece Description
0	Unused	N/A	Unused user piece size, will transmit an error and will select default user piece size.
1	Medium User Piece (default)	1.3/ 1.4/ 1.5	Medium user piece intended for ventilations between 15-180 L/min
2	Large User Piece	1.3/ 1.4/ 1.5	Large user piece intended for ventilations between 25-250 L/min
3	Unused	N/A	Unused user piece size, will transmit an error and will select default user piece size
4	RMR User Piece	1.3/ 1.4/ 1.5	RMR user piece intended for ventilations between 3-50 L/min

Get Ambient Gas Calibration Progress (Command 5)

Upon request the VO2 Master Analyzer will send an update regarding how close to completing its ambient calibration, ranging from 0-100%. If the device has exited its calibration state it will just return 100%.

Set Idle Timeout Mode (Command 11)

Sets whether the device will turn off after sitting idle. Idle is considered when the device does not have flow passing through the user piece.

Enum Value	Idle Timeout Mode	Ver.	Description
0	Idle Timeout Enabled (default)	1.3/ 1.4/ 1.5	Device will shutoff after 15 minutes of inactivity to preserve battery.
1	Idle Timeout Disabled	1.3/ 1.4/ 1.5	Device will run until the battery reaches the shut-off threshold.

Device Breath State Has Changed (Command 15)

This command is sent without any prompts from the connected application. It is used to display when the unit has detected that the user has either started to exhale or inhale through the VO2 Master Analyzer.

Enum Value	User Breath State	Ver.	Description
0	Inhale	1.3/ 1.4/ 1.5	The user has started to inhale air through the VO2 Master Analyzer
1	None	1.3/ 1.4/ 1.5	The user is neither inhaling or exhaling air.
2	Exhale	1.3/ 1.4/ 1.5	The user has started to exhale air through the VO2 Master Analyzer

Set Mask Size (Command 20)

Sets the mask size the user is wearing with the device.

Enum Value	Mask Size	Ver.	Description
0	Petite Mask	1.4/ 1.5	Testing is being performed with a petite sized mask
1	Extra Small Mask	1.4/ 1.5	Testing is being performed with an extra small sized mask
2	Small Mask	1.4/ 1.5	Testing is being performed with a small sized mask
3	Medium Mask	1.4/ 1.5	Testing is being performed with a medium sized mask
4	Large Mask	1.4/ 1.5	Testing is being performed with a large sized mask

Set Syringe Calibration Volume (Command 22)

Sets the syringe volume that is being used to calibrate the device's flow. **This command only functions on the V1.5.0 and greater versions of the device.** Volumes can vary between 0.5L and 10L and should be multiplied by 1000 before being transmitted. I.E., for a syringe volume of 5L, a value of 5000 should be transmitted.

Set Syringe Calibration Breath Count (Command 24)

Sets the number of "breaths" that are required for the flow calibration of the device using the syringe. **This command only functions on the V1.5.0 and greater versions of the device.** The number of breaths can vary between 3 and 15, with a recommendation that the number of breaths should be at least 10 for the most accurate results.

Document Revisions

Here are the changes to this document over time.

Revision	Date	Notes
V1.0.0	Oct 28, 2018	Initial version.
V2.0.0	May 28, 2019	Replace single data characteristic with one for each of Ventilation, Gas Exchange, and Environment for increased metric resolution and the addition of CO2-related parameters.
V3.0.0	Sep 2, 2020	Added VO2 Master Analyzer 1.4: widened COM IN / OUT packet size, addition of mask size setting. Removed mention of depreciated Data Characteristic.
V4.0.0	December 22, 2021	Added VO2 Master Analyzer 1.5: Addition of syringe calibration for flow measurement. Separated different versions into different headings to improve clarity.
V4.1.0	March 22, 2022	Added section to clarify the important commands sent on the COM IN/OUT characteristic.
V4.3.0	September 13, 2022	Added VO2 Master Analyzer 1.6: More stringer criteria for flow sensor calibration